

Physics LATAM - Riemannian and Complex Geometry

Homework 5 Solutions

Arsh Dhawan

Instructors: Daniel Galviz & Bowen Liu

Teaching Assistants: Anis Bousclet & Julian Alzate

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Exercises 5.1–5.8 – Solutions

Exercise 5.1

Statement: Show that any oriented Riemannian surface admits a natural almost complex structure.

Solution (expanded): Let (M, g) be a connected, oriented Riemannian surface (real dimension 2). We give two complementary constructions and show they agree.

Local orthonormal-frame construction.

1. Pick a coordinate neighbourhood $U \subset M$ and choose a local positively oriented orthonormal frame (e_1, e_2) on U , so $g(e_i, e_j) = \delta_{ij}$ and $e_1 \wedge e_2$ agrees with the chosen orientation.

2. Define a bundle endomorphism $J : TM|_U \rightarrow TM|_U$ by

$$Je_1 := e_2, \quad Je_2 := -e_1,$$

and extend $\mathbb{R}$ -linearly to all sections. For any $X = ae_1 + be_2$ we have

$$JX = aJe_1 + bJe_2 = ae_2 - be_1,$$

so a direct computation gives

$$J^2X = J(ae_2 - be_1) = a(-e_1) - b(-e_2) = -X,$$

hence $J^2 = -\text{Id}$ on U . Also for X, Y as above one checks

$$g(JX, JY) = a^2 + b^2 = g(X, Y),$$

so J is an isometry on each fibre.

3. Independence of the orthonormal frame. If (e'_1, e'_2) is another positively oriented orthonormal frame over U , then there exists a smooth function $\theta : U \rightarrow \mathbb{R}$ with

$$\begin{pmatrix} e'_1 & e'_2 \end{pmatrix} = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} e_1 & e_2 \end{pmatrix}.$$

Writing J' for the 90° rotation defined using (e'_1, e'_2) , a short matrix computation gives the commuting relation

$$R_\theta J = J R_\theta,$$

so $J' = R_\theta J R_{-\theta} = J$. Since θ depends smoothly on the base point and compositions of smooth frame transformations are smooth, the resulting global J is smooth. Thus the local definitions agree on overlaps and glue to a global smooth endomorphism J with $J^2 = -\text{Id}$ and $g(J\cdot, J\cdot) = g(\cdot, \cdot)$.

Intrinsic area-form construction.

Let ω be the area (volume) 2-form determined by g and the orientation; in any positively oriented orthonormal frame $\omega(e_1, e_2) = 1$. For each $p \in M$ define $J_p : T_p M \rightarrow T_p M$ uniquely by the identity

$$g_p(J_p X, Y) = \omega_p(X, Y) \quad (\forall X, Y \in T_p M).$$

This defines a linear map on each tangent space. Skew-symmetry of ω implies $g_p(J_p X, X) = 0$ for all X , while nondegeneracy of ω and g on a 2-dimensional space implies $J_p \neq 0$ and hence $J_p^2 = -\text{Id}$. Smooth dependence of g and ω gives smoothness of J . One checks this coincides with the orthonormal-frame construction above.

Therefore every oriented Riemannian surface admits a canonical almost complex structure J compatible with g . □

Exercise 5.2

Statement: Show that $\mathbb{C}^n$ does not have any compact complex submanifolds of positive dimension.

Solution (expanded and coordinate check): Let $X \subset \mathbb{C}^n$ be a complex submanifold of complex dimension $k \geq 1$ and suppose X is compact. Consider the smooth, real-valued function

$$\phi(z) := \|z\|^2 = \sum_{j=1}^n |z_j|^2, \quad z \in \mathbb{C}^n.$$

Compute the (1,1)-form (the Levi form) associated to ϕ in the standard coordinates: one has

$$\partial\bar{\partial}\phi = \sum_{j=1}^n dz_j \wedge d\bar{z}_j,$$

which is positive definite on complex tangent vectors (equivalently the Hermitian matrix of second derivatives is 2Id). Thus ϕ is strictly plurisubharmonic.

Restrict ϕ to X . Since X is compact, $\phi|_X$ attains a maximum at some $p \in X$. At this local maximum the complex Hessian (Levi form) of $\phi|_X$ must be seminegative in complex tangent directions, whereas the Levi form of ϕ computed above is strictly positive definite on complex directions; this contradiction shows $\phi|_X$ cannot have a local maximum unless it is locally constant. Hence $\phi|_X$ is locally constant near p , and by analyticity and connectedness it is constant on all of X . But then the Levi form of ϕ restricted to complex tangents on X would have to vanish, contradicting strict positivity. Therefore no such compact positive-dimensional complex submanifold exists.

Consequently, Stein manifolds such as $\mathbb{C}^n$ contain no compact positive-dimensional complex subvarieties. □

Exercise 5.3

Statement: Show that any holomorphic map from $\mathbb{C}P^1$ into a complex torus is constant. What about maps from $\mathbb{C}P^n$ into a complex torus?

Solution (concise): Let $T = \mathbb{C}^m/\Lambda$ be a complex torus and let $f : \mathbb{C}P^n \rightarrow T$ be holomorphic. Since $\mathbb{C}P^n$ is simply connected, f lifts to a holomorphic map $F : \mathbb{C}P^n \rightarrow \mathbb{C}^m$ with $f = \pi \circ F$ where π is the quotient. But each component of F is a holomorphic function on the compact manifold $\mathbb{C}P^n$, hence constant; thus F (and therefore f) is constant. □

Exercise 5.4

Statement: Check that the Plücker embedding is injective. As a consequence, the Grassmannian manifold is projective.

Solution (expanded): Let $V, W \subset \mathbb{C}^n$ be k -dimensional subspaces. Pick bases $v_1, \dots, v_k$ of V and $w_1, \dots, w_k$ of W . The Plücker map sends V to the projective class $[v_1 \wedge \dots \wedge v_k]$ in $\mathbb{P}(\wedge^k \mathbb{C}^n)$. Suppose

$$v_1 \wedge \dots \wedge v_k = \lambda w_1 \wedge \dots \wedge w_k \quad (\lambda \in \mathbb{C}^*).$$

Choose vectors $u_1, \dots, u_{n-k}$ so that $w_1, \dots, w_k, u_1, \dots, u_{n-k}$ is a basis of $\mathbb{C}^n$. Wedge both sides with $u_1 \wedge \dots \wedge u_{n-k}$ to obtain

$$v_1 \wedge \dots \wedge v_k \wedge u_1 \wedge \dots \wedge u_{n-k} = \lambda w_1 \wedge \dots \wedge w_k \wedge u_1 \wedge \dots \wedge u_{n-k}.$$

The right-hand side is a nonzero top-form (since the w_i and u_j form a basis), hence the left-hand side is nonzero, which implies the vectors $v_1, \dots, v_k$ lie in the span of $w_1, \dots, w_k$. Thus $V \subset W$. Reversing the roles shows $W \subset V$, so $V = W$. This proves injectivity.

Algebraically, write the v_i and w_j as rows of $k \times n$ matrices; equality of all Plücker coordinates (the $k \times k$ minors) implies row spaces coincide, another way to see $V = W$. The image is defined by the quadratic Plücker relations, hence projective. $\square$

Exercise 5.5

Statement: For $\mathbb{C}^n \cong \mathbb{R}^{2n}$ with coordinates $z_j = x_j + iy_j$, show the relation between the real volume form and the complex form.

Solution (detailed): Write $z_j = x_j + iy_j$ and $\bar{z}_j = x_j - iy_j$. Then

$$dz_j = dx_j + i dy_j, \quad d\bar{z}_j = dx_j - i dy_j.$$

Compute the wedge product for a single index (expand and cancel the $dx_j \wedge dx_j$ and $dy_j \wedge dy_j$ terms):

$$\begin{aligned} dz_j \wedge d\bar{z}_j &= (dx_j + i dy_j) \wedge (dx_j - i dy_j) \\ &= dx_j \wedge dx_j - i dx_j \wedge dy_j + i dy_j \wedge dx_j + (-i)(-i) dy_j \wedge dy_j \\ &= -2i dx_j \wedge dy_j, \end{aligned}$$

because $dx_j \wedge dx_j = dy_j \wedge dy_j = 0$ and $dy_j \wedge dx_j = -dx_j \wedge dy_j$.

Wedgeing over $j = 1, \dots, n$ yields

$$\bigwedge_{j=1}^n dz_j \wedge d\bar{z}_j = (-2i)^n \bigwedge_{j=1}^n dx_j \wedge dy_j,$$

and the displayed equivalence follows. $\square$

Exercise 5.6

Statement: Let X be a complex manifold. Prove that $(T_{\mathbb{R}}X, J)$ is isomorphic to $T^{1,0}X$ as complex vector bundles.

Solution (expanded with inverse): Complexify the real tangent bundle to obtain

$$T_{\mathbb{R}}X \otimes \mathbb{C} = T^{1,0}X \oplus T^{0,1}X,$$

where $T^{1,0}X$ (resp. $T^{0,1}X$) is the $+i$ (resp. $-i$) eigensubbundle of J extended complex-linearly. Define the projection onto $T^{1,0}X$ by

$$P^{1,0} = \frac{1}{2}(\text{Id} - iJ) : T_{\mathbb{R}}X \otimes \mathbb{C} \rightarrow T^{1,0}X.$$

Restricted to $T_{\mathbb{R}}X$ this gives a $\mathbb{C}$ -linear map (when $T_{\mathbb{R}}X$ is regarded as a complex bundle via J)

$$\Phi(v) := P^{1,0}(v) = \frac{1}{2}(v - iJv) \in T_x^{1,0}X.$$

Note that if $w \in T_x^{1,0}X$ then $Jw = iw$, so

$$P^{1,0}(w) = \frac{1}{2}(w - iJw) = \frac{1}{2}(w - i(iw)) = w,$$

while $P^{1,0}$ annihilates the $-i$ -eigensubbundle $T^{0,1}X$, so $P^{1,0}$ is indeed the projection onto the $+i$ eigenspace. To see Φ is an isomorphism of complex bundles, construct the inverse on $T^{1,0}X$: for $w \in T_x^{1,0}X$ write $w = u - iJu$ with $u \in T_{\mathbb{R}}X$ (taking real and imaginary parts). Then $\Phi(u) = w$, so Φ is surjective and fiberwise injective; equal complex ranks force it to be an isomorphism. Thus $(T_{\mathbb{R}}X, J) \cong T^{1,0}X$ as complex vector bundles. $\square$

Exercise 5.7

Statement: Are there holomorphic vector fields on $\mathbb{C}P^n$ which vanish only in a finite number of points? If yes, in how many?

Solution (expanded): The space of global holomorphic vector fields on $\mathbb{C}P^n$ is the Lie algebra of holomorphic automorphisms, isomorphic to $\mathfrak{sl}(n+1, \mathbb{C})$ modulo scalars; hence

$$\dim H^0(\mathbb{C}P^n, T\mathbb{C}P^n) = (n+1)^2 - 1.$$

If a holomorphic vector field V has only isolated zeros, the holomorphic Poincaré–Hopf theorem (a special case of the Bott residue theorem) states that the sum of the local indices of the zeros equals the topological Euler characteristic of $\mathbb{C}P^n$, namely $\chi(\mathbb{C}P^n) = n+1$. Thus the total number of zeros counted with multiplicity is $n+1$.

Example ($\mathbb{C}P^1$): on the Riemann sphere holomorphic vector fields are global sections of $T\mathbb{C}P^1 \cong \mathcal{O}(2)$ and are given in an affine coordinate z by $(a + bz + cz^2)\partial/\partial z$, which generically have two zeros (counted with multiplicity), matching $\chi = 2$. $\square$

Exercise 5.8

Statement: Prove the Euler sequence on $\mathbb{C}P^n$, i.e. the exact sequence

$$0 \rightarrow \mathcal{O}_{\mathbb{C}P^n} \xrightarrow{s} \mathcal{O}_{\mathbb{C}P^n}(1)^{\oplus(n+1)} \xrightarrow{\rho} T\mathbb{C}P^n \rightarrow 0.$$

Solution (with local chart verification): Let $\pi : \mathbb{C}^{n+1} \setminus \{0\} \rightarrow \mathbb{C}P^n$ be the quotient. The tautological subbundle $\mathcal{O}(-1)$ sits inside the trivial bundle $\mathbb{C}^{n+1} \times (\mathbb{C}^{n+1} \setminus \{0\})$ with fibre over $[Z]$ equal to the line spanned by Z ; its dual is $\mathcal{O}(1)$. Points of $\mathcal{O}(1)^{\oplus(n+1)}$ are therefore $(n+1)$ -tuples of homogeneous linear functions.

Define

$$s : \mathcal{O} \rightarrow \mathcal{O}(1)^{\oplus(n+1)}, \quad 1 \mapsto (Z_0, \dots, Z_n),$$

and define ρ by sending an $(n+1)$ -tuple of linear functions $(t_0, \dots, t_n)$ to the induced tangent vector on projective space obtained by differentiating the linear action on $\mathbb{C}^{n+1}$ and descending modulo the Euler (radial) vector field. Concretely one may write

$$\rho(t_0, \dots, t_n) = \sum_{i=0}^n t_i \frac{\partial}{\partial Z_i} \quad (\text{viewed modulo the Euler vector field } \sum_i Z_i \partial / \partial Z_i),$$

so two tuples that differ by a multiple of $(Z_0, \dots, Z_n)$ define the same tangent vector on projective space.

To check exactness explicitly on the affine chart $U_0 = \{Z_0 \neq 0\}$ set $z_i = Z_i/Z_0$ ($i = 1, \dots, n$). A local section of $\mathcal{O}(1)^{\oplus(n+1)}$ corresponds to an $(n+1)$ -tuple of functions $(a_0(Z), \dots, a_n(Z))$ homogeneous of degree 1; dividing by Z_0 yields a tuple of functions on U_0 which we view as an element of $\mathbb{C}^{n+1}$. The projection ρ becomes the linear map

$$\mathbb{C}^{n+1} \rightarrow \mathbb{C}^n, \quad (t_0, t_1, \dots, t_n) \mapsto (t_1 - t_0 z_1, \dots, t_n - t_0 z_n),$$

whose kernel is precisely the 1-dimensional subspace spanned by $(1, z_1, \dots, z_n)$, i.e. the image of s . Thus the sequence is exact on U_0 ; identical checks on other charts and compatibility on overlaps yield global exactness of locally free sheaves, i.e. the Euler sequence.

Algebraically the kernel is the Euler vector (scalars), and the cokernel is the tangent bundle. This completes the verification of the Euler sequence. $\square$